

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 - CONCEPT MAPPING

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	C01	Assessment:	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	Total Marks:	25
CO1 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
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Department:	CSE (AI/ML)	Date:	05/08/2026

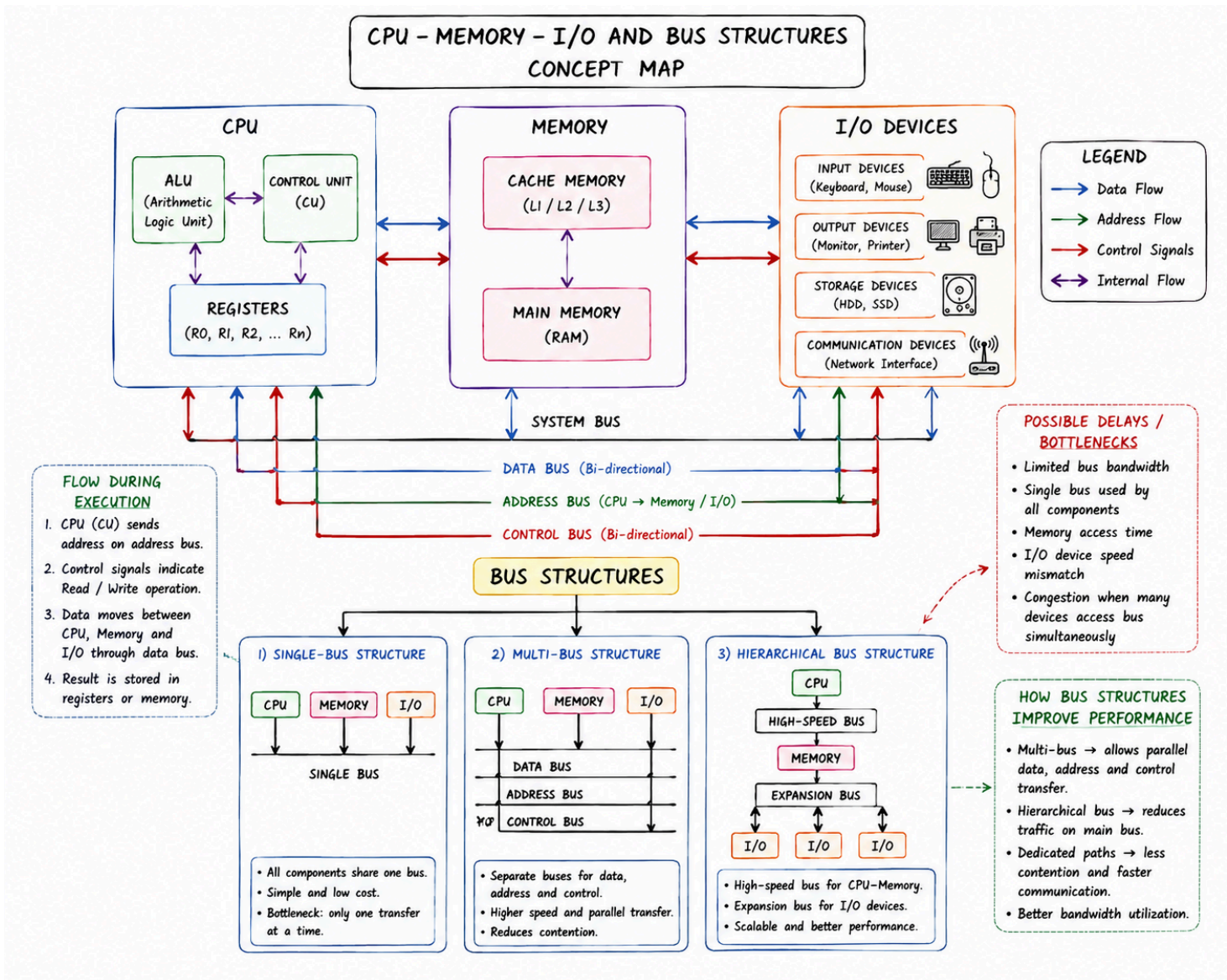
Instruction to Students

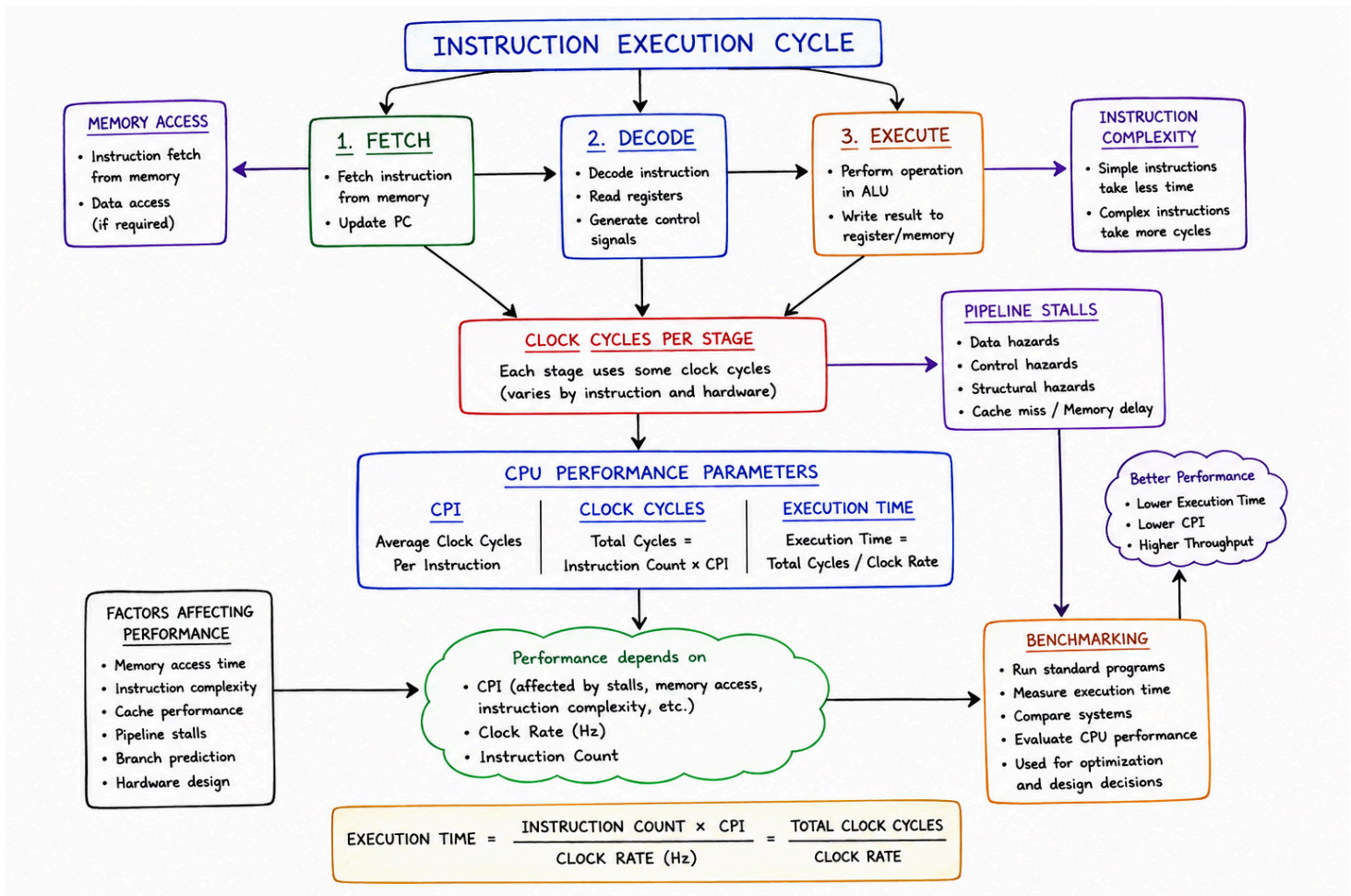
- Answer two questions as per the Instruction.
- Each question carries 12.5 mark.
- Only the appropriate Tools can be used
- Time allotted for the exam is 60 min

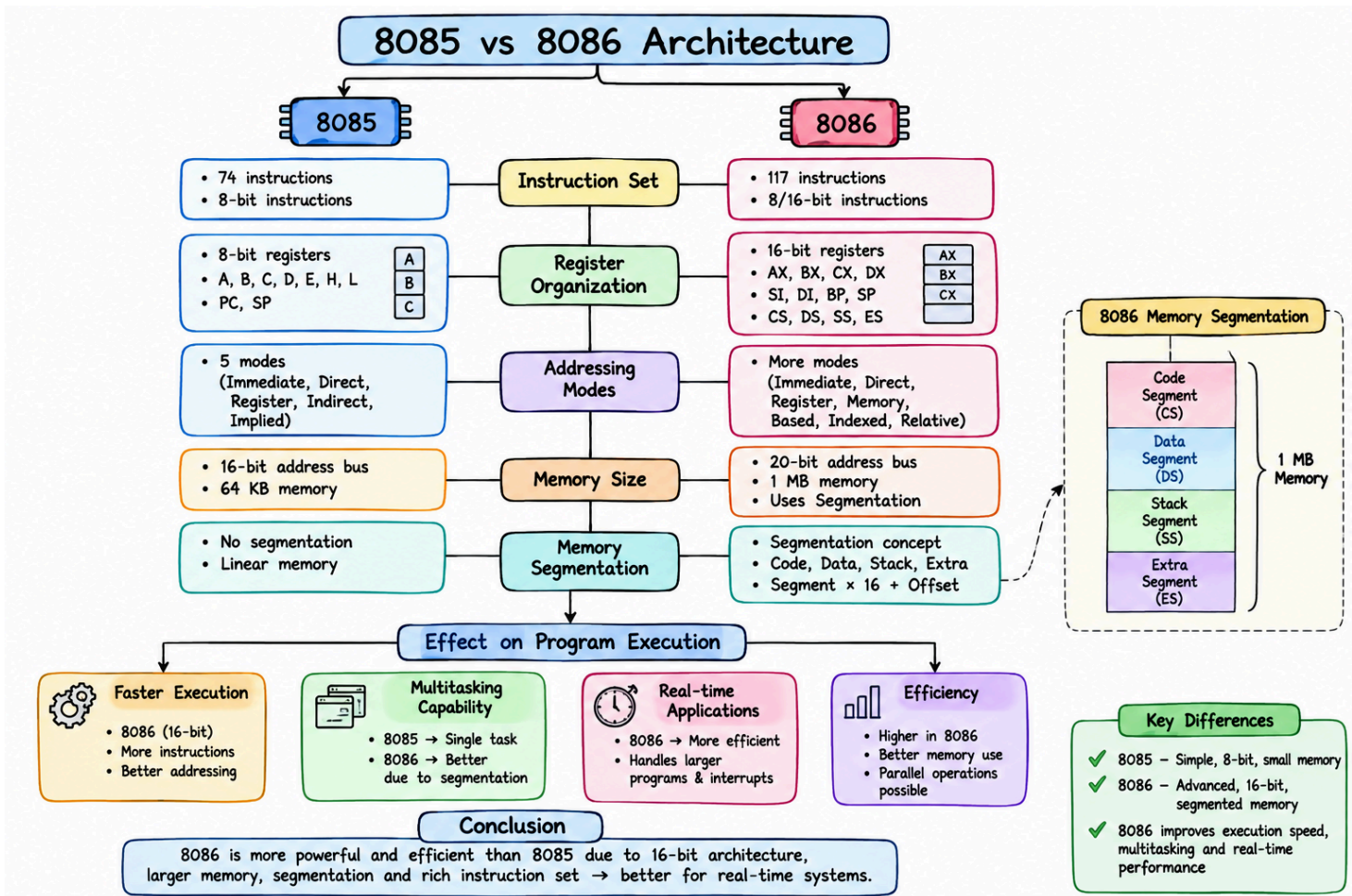
ANNEXURE A

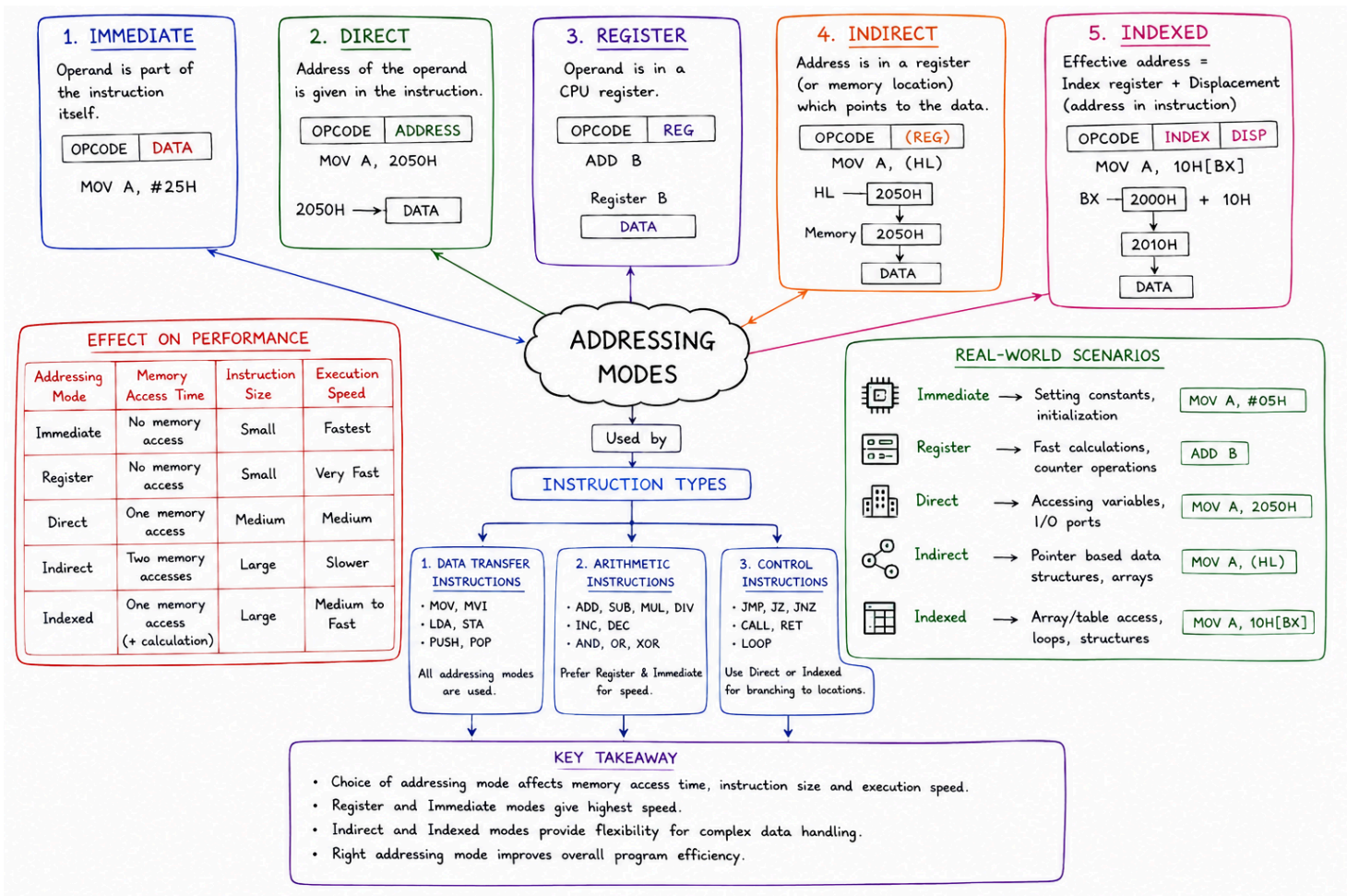
1. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.
2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.
3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.
4. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show real-world scenarios where specific addressing modes are preferred.
5. Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system des

1.









NUMBER SYSTEMS, INSTRUCTIONS, MEMORY AND DEBUGGING

1. NUMBER SYSTEMS

BINARY (Base 2)

- Uses 0 and 1
- Used by hardware internally
- Each bit = smallest unit of data

HEXADECIMAL (Base 16)

- Uses 0-9, A-F
- Shorthand for binary
- 1 Hex digit = 4 bits

BINARY ↔ HEXADECIMAL

0000	0001	0010	0011	0100	0101	0110	0111
0	1	2	3	4	5	6	7
1000	1001	1010	1011	1100	1101	1110	1111
8	9	A	B	C	D	E	F

2. DATA CONVERSION

Decimal → Binary

$$25_{10} = 11001_2$$

Binary → Hex

$$11001_2 = 0001\ 1001_2 = 19_{16}$$

Hex → Binary

$$3A_{16} = 0011\ 1010_2$$

Hex → Decimal

$$3A_{16} = (3 \times 16) + 10 = 58_{10}$$

3. INSTRUCTION REPRESENTATION

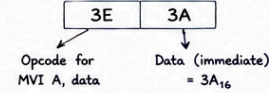
Instruction Format

OPCODE | OPERAND(s) | MODE

- OPCODE → operation to perform
- OPERAND(s) → data or address
- MODE → addressing mode info

Example (8085)

MVI A, 3AH



4. MEMORY STORAGE

- Memory stores everything in BINARY.
- Hex is used to represent memory contents in a compact form.

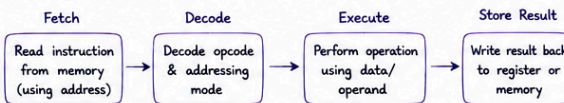
Example Memory (starting at 2000H)

Address (HEX)	Data (HEX)	Data (Binary)
2000	3E	0011 1110
2001	3A	0011 1010
2002	32	0011 0010
2003	10	0001 0000

Flow

- CPU uses address bus to read instruction.
- Instruction is in binary.
- CPU decodes and executes it.

5. CPU FETCH - DECODE - EXECUTE CYCLE



6. WHY HEXADECIMAL?

- Easier for humans to read than long binary.
- Compact representation (1 hex digit = 4 bits).
- Widely used in assembly language, memory dumps, machine code, system design.

7. DEBUGGING PROCESS

1. View Memory



Memory contents shown in HEX.

2. Interpret Instruction



Convert hex to binary (if needed) and understand opcode and operands.

3. Check Execution



Verify if instruction executes as expected.

4. Identify Error



Find wrong opcode, wrong data, or incorrect address.

5. Fix and Test



Correct the code/data and test again.

QUICK REFERENCE

- 1 Byte = 8 bits = 2 Hex digits
- Example:
 $1010\ 1101_2 = AD_{16}$
 $AD_{16} = 1010\ 1101_2$
- Word (16-bit) = 4 Hex digits

SUMMARY

Binary is the language of computers. Hexadecimal is the language of humans dealing with computers. Hex makes instruction representation, memory viewing, and debugging simpler and efficient.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand